Arpit Singh

M.Tech Artificial Intelligence & Machine Learning - TA

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Degree	University/Institute	Year	CGPA/Marks(%)
M.Tech Artificial Intelligence & Machine Learning - TA	IIT Hyderabad	2027	9.25
COMPUTER SCIENCE WITH MAJORS IN AI	IIITDM KANCHEEPURAM	2025	7.94
XII (CBSE)	BETHEL ACADEMY ,PRAYAGRAJ	2021	83.80%
X (CBSE)	ARMY PUBLIC SCHOOL ,PRAYAGRAJ	2019	97.40%

POSITIONS OF RESPONSIBILITY

1. Teaching Assistant for Practical Challenges of Image Analysis (Jan 2026- May2026)
2. Teaching assistant for Fundamentals of Machine Learning (Aug 2026-Nov 2026)

SKILLS

RAG , LLM , NLP , Langsmith , Langchain , Langgraph , API , C ,C++ , Python , SQL , Streamlit , Docker , Kubernetes , Pytorch , TensorFlow , Time Series Data

RELEVANT COURSES

Fundamentals of Machine Learning , Natural Language Processing ,Deep Learning , Statistical Inference, Probability and Statistics, Convex Optimization , Advanced Data Structures , Linear Algebra .

PROJECTS

Company Evaluation System

1. Deployed using **Tavily** Api for getting the results from the web , **Langchain** for the reasoning and the summarisation process and **Streamlit** for the deployment of the project , with **Docker** for containerisation of the application
2. This project tries to get the results , assets , projects , debt , acquisitions of the companies , so to make it easy for the investors to get an idea of the companies **SWOT** analysis , it also summarizes the results to get the overall view of the company .

Customer Assistant

Built a Customer Assistant for assisting customers with their orders . Created using **RAG** for processing the documents) , vector store namely **ChromaDB** , for getting the documents related to our query , **Langchain**(for handling conversations with the customer and then pass it to a human for further redressal) and **Streamlit** (for frontend) .This was then **Dockerized** . For converting into images , which can then be distributed

1. They can place order through the site .
2. They can post their feedback in the site itself , where the task orchestrator does the work of routing the request to different subroutines .
3. Be it for getting the feedback or to know the company policy , the task orchestrator along with the OLLAMA -miniLm handle the task .
4. For maintaining the good context , the documents are not blindly spliced , instead , we used the **Semantic Chunker** so that the parts with the same meaning are not separated , and the hence the context is not diluted . We then embed the query as well these semantically separated docs , where the **RAG** system then gets the document which are relatable to the query .

EXTRACURRICULAR

1. Badminton (School level Badminton Player twice in a row)
2. Gymnasium

INTERNSHIPS

IIT BOMBAY

1. Worked on the problem of **Scene Understanding** which is still an open ended problem of **Computer Vision.**
2. Worked on **Semantic Segmentation** and tried using **attention methods** on the readily available models .
3. We also worked on idea of **Knowledge distillation** , which is used for distilling the information from a big model to a smaller model .
4. Worked on **CBAM** and **BAM** attention modules for distilling the knowledge into a smaller model , while not leading to a drastic decrease in the accuracy of the model